

Proofs and Properties of Equality and Congruence Homework

Prove the following statements using two-column proofs.

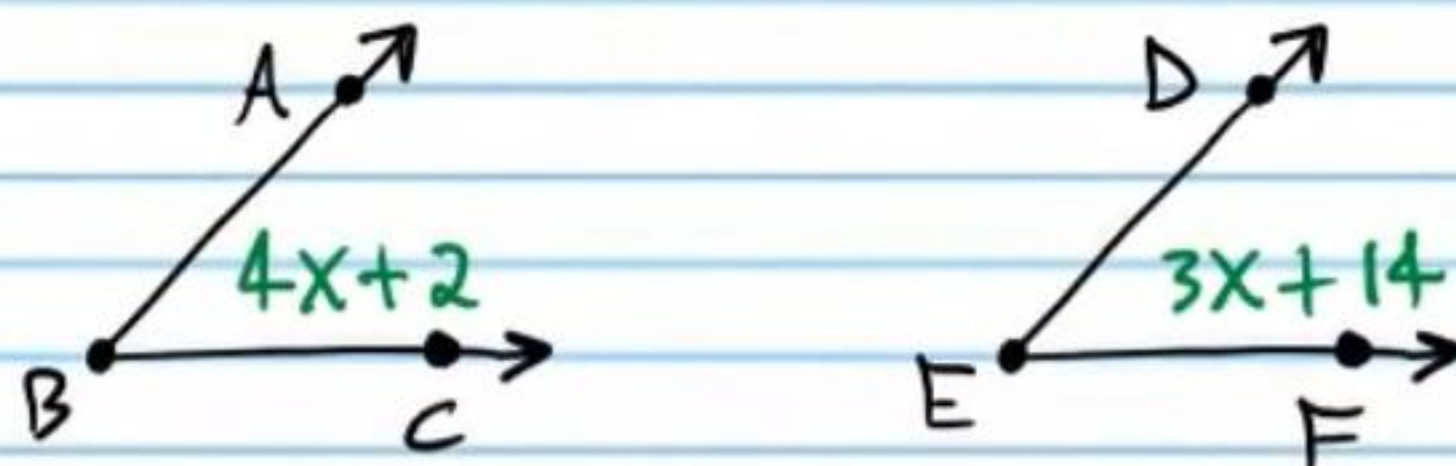
- ① If the two segments below are congruent, then $x = 2$.



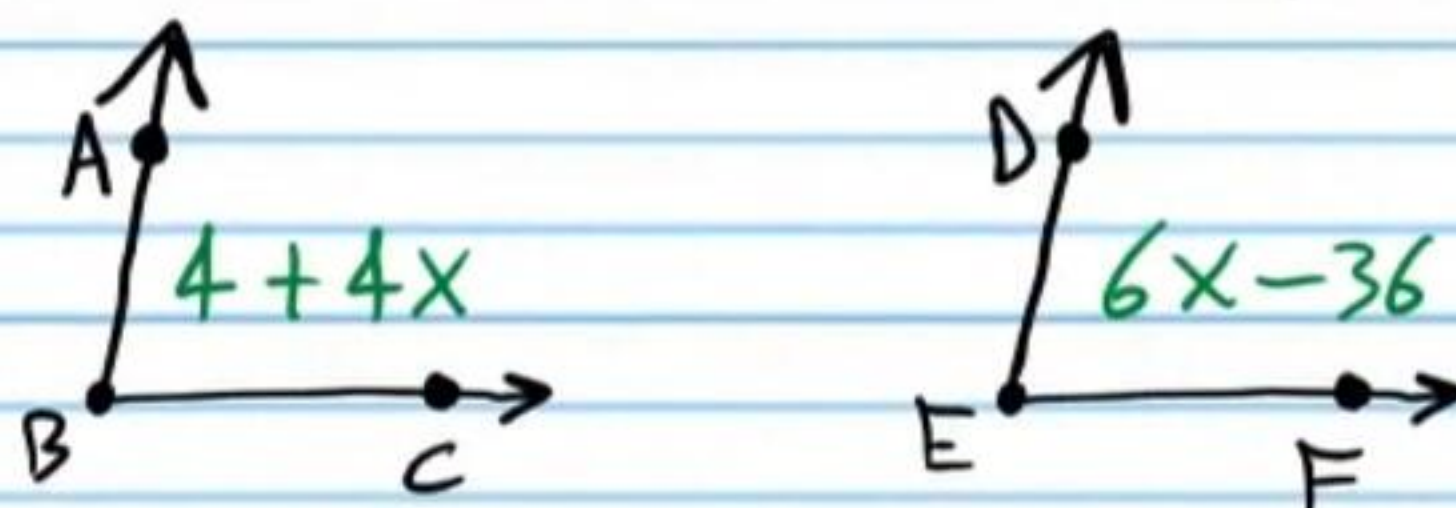
- ② If the two segments below are congruent, then $AB = 40$.



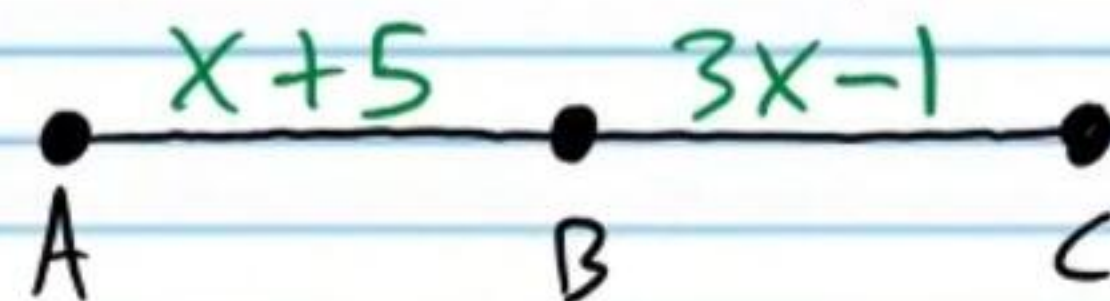
- ③ If the two angles below are congruent, then $x = 12$.



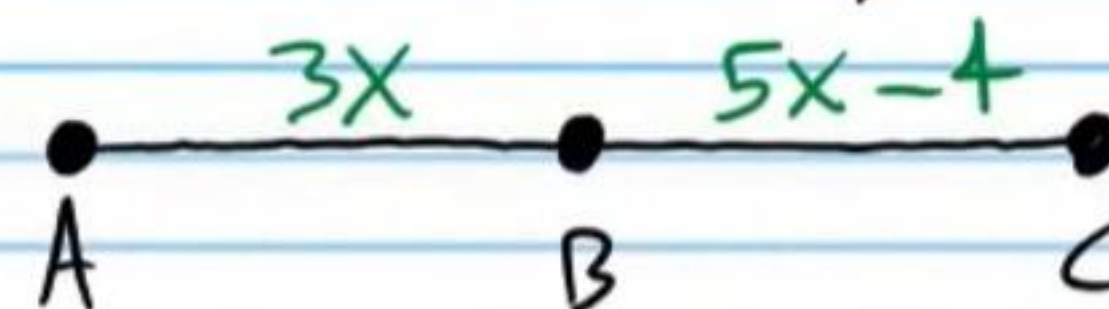
- ④ If the two angles below are congruent, then $m\angle DEF = 84^\circ$.



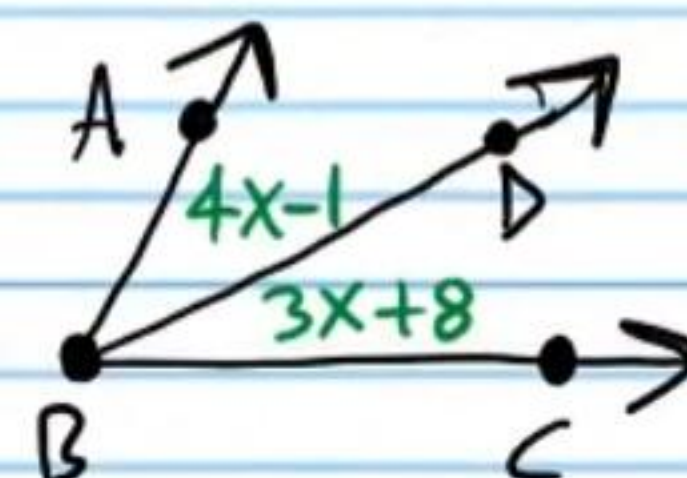
- ⑤ If point B bisects $\overline{AC}$, then $x = 3$.



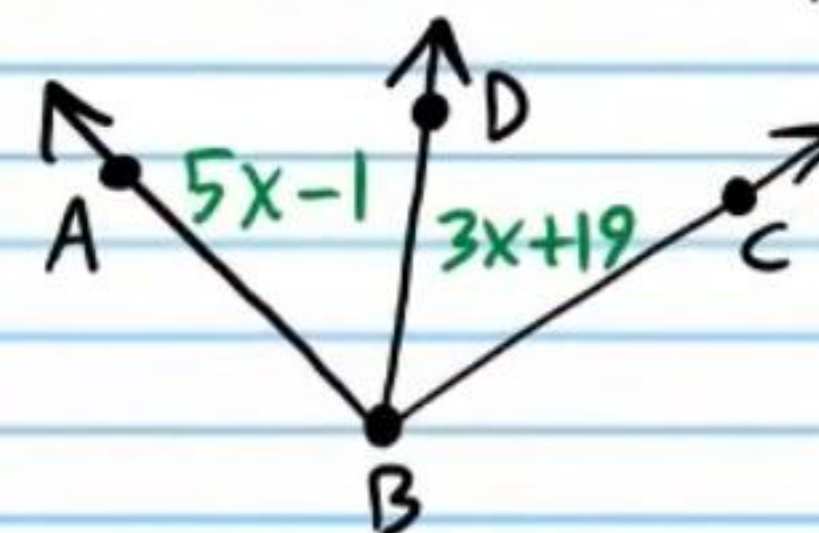
- ⑥ If point B bisects $\overline{AC}$, then $BC = 6$.



- ⑦ If $\overrightarrow{BD}$ bisects $\angle ABC$, then $x = 9$.



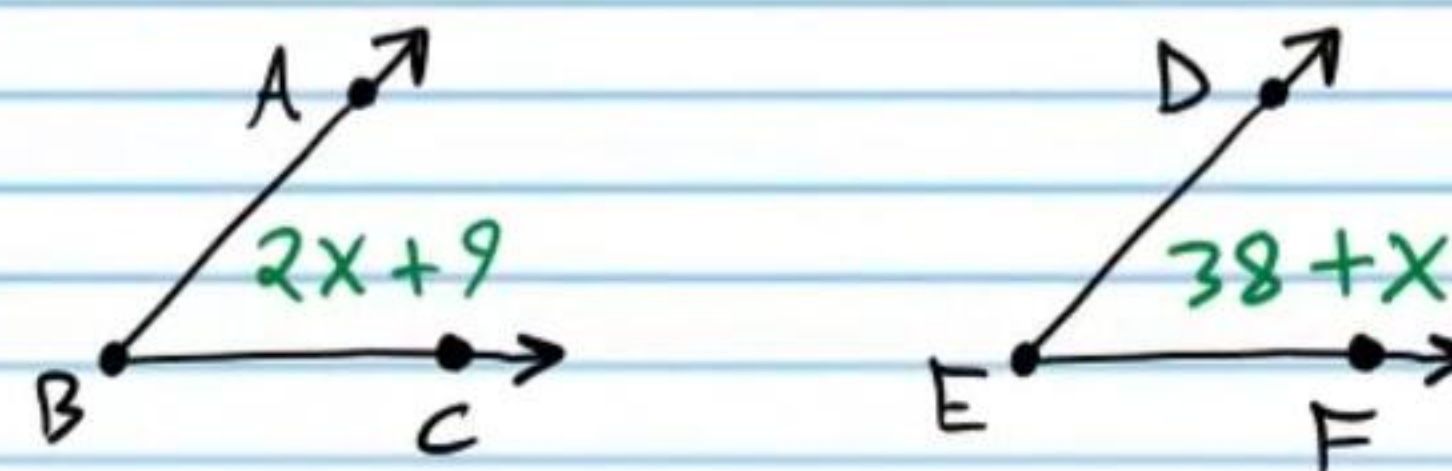
- ⑧ If $\overrightarrow{BD}$ bisects $\angle ABC$, then $m\angle DBC = 49^\circ$.



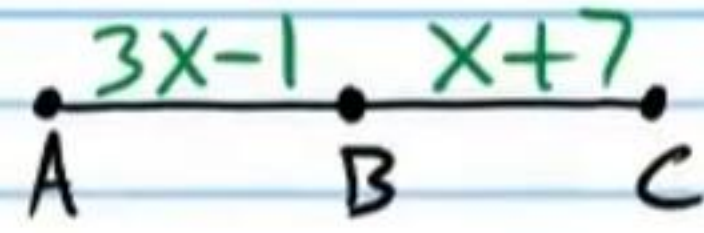
- ⑨ If the two segments below are congruent, then $x = 10$.



- ⑩ If the two angles below are congruent, then $x = 29$.



⑪ If point B bisects $\overline{AC}$, then $AB = 11$.



⑫ The converse of problem #1.

⑬ The converse of problem #4.

⑭ The converse of problem #6.

⑮ The converse of problem #7.

Answers

①	Statement	Reason
I)	$\overline{AB} \cong \overline{CD}$	Given.
II)	$AB = CD$	Def. of $\cong$ Seg.
III)	$AB = 2x+1$ & $CD = 9-2x$	Given (diagram)
IV)	$2x+1 = 9-2x$	Subs. Prop of =.
V)	$4x+1 = 9$	Add Prop of =.
VI)	$4x = 8$	Subt. Prop of =.
VII)	$x = 2$	Div. Prop of =.

②	Statement	Reason
I)	$\overline{AB} \cong \overline{CD}$	Given.
II)	$AB = CD$	Def. of $\cong$ Seg.
III)	$AB = 5x+5$ & $CD = -3x+61$	Given (diagram)
IV)	$5x+5 = -3x+61$	Subs. Prop of =.
V)	$8x+5 = 61$	Add Prop of =.
VI)	$8x = 56$	Subt. Prop of =.
VII)	$x = 7$	Div. Prop of =.

VIII)	$AB = 5(7)+5$	Subs. Prop of =.
IX)	$AB = 40$	Simplification.

③	Statement	Reason
I)	$\angle ABC \cong \angle DEF$	Given.
II)	$m\angle ABC = m\angle DEF$	Def. of $\cong$ $\angle$'s.
III)	$m\angle ABC = 4x+2$ and $m\angle DEF = 3x+14$.	Given (diagram).
IV)	$4x+2 = 3x+14$	Subs. Prop of =.
V)	$x+2 = 14$	Subt. Prop of =.
VI)	$x = 12$	Subt. Prop of =.

④	Statement	Reason
I)	$\angle ABC \cong \angle DEF$	Given.
II)	$m\angle ABC = m\angle DEF$	Def. of $\cong$ $\angle$'s.
III)	$m\angle ABC = 4+4x$ and $m\angle DEF = 6x-36$	Given (diagram).
IV)	$4+4x = 6x-36$	Subs. Prop of =.

V) $4 = 2x - 36$	Subst. Prop of =.
VI) $40 = 2x$	Add. Prop of =.
VII) $20 = x$	Div. Prop of =.
VIII) $m\angle DEF = 6(20) - 36$	Subst. Prop of =.
IX) $m\angle DEF = 84^\circ$	Simplification.

⑤

Statement	Reason
I) B bisects $\overline{AC}$.	Given.
II) $AB = BC$	Def. of seg bisector.
III) $AB = x + 5$ and $BC = 3x - 1$	Given (diagram).
IV) $x + 5 = 3x - 1$	Subst. Prop of =.
V) $5 = 2x - 1$	Subst. Prop of =.
VI) $6 = 2x$	Add Prop of =.
VII) $3 = x$	Div Prop of =.
VIII) $x = 3$	Symm. Prop of =.

⑥

Statement	Reason
I) B bisects $\overline{AC}$.	Given.

II) $AB = BC$	Def. of seg bisector.
III) $AB = 3x$ and $BC = 5x - 4$	Given (diagram).
IV) $3x = 5x - 4$	Subst. Prop of =.
V) $-2x = -4$	Subst. Prop of =.
VI) $x = 2$	Div Prop of =.
VII) $BC = 5(2) - 4$	Subst. Prop of =.
VIII) $BC = 6$	Simplification

⑦

Statement	Reason
I) $\overrightarrow{BD}$ bisects $\angle ABC$.	Given.
II) $m\angle ABD = m\angle DBC$	Def. of $\angle$ bisector.
III) $m\angle ABD = 4x - 1$ and $m\angle DBC = 3x + 8$.	Given (diagram)
IV) $4x - 1 = 3x + 8$	Subst. Prop of =.
V) $x - 1 = 8$	Subst. Prop of =.
VI) $x = 9$	Add Prop of =.

⑧

Statement	Reason
I) $\overrightarrow{BD}$ bisects $\angle ABC$.	Given.
II) $m\angle ABD = m\angle DBC$	Def. of $\angle$ bisector.
III) $m\angle ABD = 5x - 1$ and $m\angle DBC = 3x + 19$.	Given (diagram)
IV) $5x - 1 = 3x + 19$	Subs. Prop of =.
V) $2x - 1 = 19$	Subt. Prop of =.
VI) $2x = 20$	Add Prop of =.
VII) $x = 10$	Div. Prop of =.
VIII) $m\angle DBC = 3(10) + 19$	Subs. Prop of =.
IX) $m\angle DBC = 49^\circ$	Simplification.

⑨

Statement	Reason
I) $\overline{AB} \cong \overline{CD}$	Given.
II) $AB = CD$	Def. of $\cong$ seg.
III) $AB = 7x + 5$ & $CD = 9x - 15$	Given (diagram)
IV) $7x + 5 = 9x - 15$	Subs. Prop of =.

V) $5 = 2x - 15$	Subt. Prop of =.
VI) $20 = 2x$	Add Prop of =.
VII) $10 = x$	Div. Prop of =.
VIII) $x = 10$	Symm. Prop of =.

⑩

Statement	Reason
I) $\angle ABC \cong \angle DEF$	Given.
II) $m\angle ABC = m\angle DEF$	Def. of $\cong$ $\angle$'s.
III) $m\angle ABC = 2x + 9$ and $m\angle DEF = 38 + x$.	Given (diagram).
IV) $2x + 9 = 38 + x$	Subs. Prop of =.
V) $x + 9 = 38$	Subt. Prop of =.
VI) $x = 29$	Subt. Prop of =.

⑪

Statement	Reason
I) B bisects $\overline{AC}$.	Given.
II) $AB = BC$	Def. of seg bisector.
III) $AB = 3x - 1$ and $BC = x + 7$	Given (diagram).
IV) $3x - 1 = x + 7$	Subs. Prop of =.

V) $2x-1=7$	Subl. Prop of =.
VI) $2x=8$	Add Prop of =.
VII) $x=4$	Div Prop of =.
VIII) $AB=3(4)-1$	Subs. Prop of =.
IX) $AB=11$	Simplification.

(12)

Statement	Reason
I) $x=2$	Given.
II) $AB=2x+1$ & $CD=9-2x$	Given (diagram).
III) $AB=2(2)+1$ & $CD=9-2(2)$	Subs. Prop of =.
IV) $AB=5$ & $CD=5$	Simplification
V) $AB=CD$	Trans. Prop of =.
VI) $\overline{AB} \cong \overline{CD}$	Def. of $\cong$ seg

(13)

Statement	Reason
I) $m\angle DEF = 84^\circ$	Given.
II) $m\angle DEF = 6x-36$	Given (diagram)

III) $84=6x-36$	Trans. Prop of =.
IV) $120=6x$	Add Prop of =.
V) $20=x$	Div Prop of =.
VI) $m\angle ABC = 4+4x$	Given (diagram)
VII) $m\angle ABC = 4+4(20)$	Subs. Prop of =.
VIII) $m\angle ABC = 84^\circ$	Simplification.
IX) $m\angle ABC = m\angle DEF$	Trans. Prop of =.
X) $\angle ABC \cong \angle DEF$	Def. of $\cong$ $\angle$'s.

(14)

Statement	Reason
I) $BC=6$	Given.
II) $BC=5x-4$	Given (diagram)
III) $6=5x-4$	Trans. Prop of =.
IV) $10=5x$	Add Prop of =.
V) $2=x$	Div. Prop of =.
VI) $AB=3x$	Given (diagram)
VII) $AB=3(2)$	Subs. Prop of =.
VIII) $AB=6$	Simplification.

$$\text{IX) } AB = BC$$

Trans. Prop of =.

X) B bisects $\overline{AC}$.

Def. of seg bisector.

(15)

Statement	Reason
I) $x = 9$	Given
II) $m\angle ABD = 4x - 1$ and $m\angle DBC = 3x + 8$	Given (diagram)
III) $m\angle ABD = 4(9) - 1$ and $m\angle DBC = 3(9) + 8$	Subs. Prop of =.
IV) $m\angle ABD = 35^\circ$ and $m\angle DBC = 35^\circ$	Simplification.
V) $m\angle ABD = m\angle DBC$	Trans. Prop of =.
VI) $\overrightarrow{BD}$ bisects $\angle ABC$	Def. of $\angle$ bisector.